

## VILJANDI, Estonia

WMO: 262330

Lat: 58.3778N

Lon: 25.6003E

Elev: 288

StdP: 14.54

Time Zone: 2.00 (EUE)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -6.9       | 0.1 | -11.1                         | 3.1 | -6.4 | -4.3 | 4.4 | 1.0  | 16.2                  | 35.3 | 14.5 | 33.5 | 2.9                   | 60   | 0.537 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.5                   | 82.3            | 67.3 | 78.9 | 64.9 | 75.7 | 63.6 | 69.2                | 78.2 | 67.1 | 75.5 | 65.2 | 73.2 | 5.1                  | 80   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 66.0                            | 96.9 | 72.9 | 64.0 | 90.1 | 71.2 | 62.0 | 84.1 | 69.1 | 33.4          | 78.3 | 31.7 | 75.3 | 30.3 | 73.3 | 76.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 13.6              | 11.9 | 10.5 | -11.2                      | 86.5 | 8.7                | 3.5 | -17.5                                              | 89.0 | -22.6      | 91.1 | -27.5      | 93.1 | -33.8      | 95.6 |
|                   |      |      | -11.2                      | 71.9 | 8.1                | 1.9 | -17.1                                              | 73.3 | -21.8      | 74.4 | -26.4      | 75.5 | -32.3      | 76.9 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 43.2        | 22.8  | 23.3  | 30.5  | 41.6  | 53.0  | 59.3  | 64.6  | 62.2  | 53.6  | 42.5  | 34.9  | 29.0  | (6)  |
|                                                                        | DBStd       | 16.54       | 11.71 | 11.13 | 8.36  | 6.83  | 7.51  | 5.92  | 5.52  | 5.23  | 5.64  | 7.37  | 7.30  | 9.68  | (7)  |
|                                                                        | HDD50       | 3892        | 842   | 749   | 604   | 263   | 51    | 4     | 0     | 30    | 247   | 452   | 650   |       | (8)  |
|                                                                        | HDD65       | 8062        | 1307  | 1169  | 1069  | 701   | 378   | 189   | 75    | 118   | 342   | 697   | 902   | 1115  | (9)  |
|                                                                        | CDD50       | 1426        | 0     | 0     | 0     | 12    | 145   | 283   | 453   | 380   | 138   | 15    | 0     | 0     | (10) |
|                                                                        | CDD65       | 121         | 0     | 0     | 0     | 0     | 7     | 18    | 64    | 32    | 0     | 0     | 0     | 0     | (11) |
|                                                                        | CDH74       | 1034        | 0     | 0     | 0     | 0     | 81    | 183   | 517   | 250   | 3     | 0     | 0     | 0     | (12) |
|                                                                        | CDH80       | 190         | 0     | 0     | 0     | 0     | 8     | 27    | 110   | 45    | 0     | 0     | 0     | 0     | (13) |
| (14) Wind                                                              | WSAvg       | 5.0         | 5.6   | 5.4   | 5.5   | 5.1   | 4.6   | 4.6   | 4.0   | 4.0   | 4.1   | 4.9   | 5.6   | 6.3   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 22.1        | 1.5   | 1.1   | 1.1   | 1.1   | 1.1   | 1.4   | 1.9   | 2.4   | 2.6   | 2.5   | 2.7   | 2.1   | (15) |
|                                                                        | PrecMax     | 30.3        | 3.0   | 2.6   | 2.5   | 2.8   | 3.4   | 3.7   | 4.6   | 7.2   | 6.4   | 6.1   | 5.2   | 4.4   | (16) |
|                                                                        | PrecMin     | 16.3        | 0.2   | 0.0   | 0.1   | 0.0   | 0.2   | 0.0   | 0.6   | 0.0   | 0.8   | 0.5   | 0.7   | 0.2   | (17) |
|                                                                        | PrecStd     | 3.3         | 0.7   | 0.7   | 0.5   | 0.7   | 0.7   | 0.9   | 1.1   | 1.5   | 1.3   | 1.3   | 1.0   | 0.9   | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 43.2  | 42.3  | 56.3  | 69.6  | 80.7  | 83.5  | 86.4  | 84.8  | 73.1  | 60.2  | 49.6  | 47.8  | (19) |
|                                                                        |             | MCWB        | 41.8  | 39.1  | 44.0  | 51.3  | 62.1  | 66.3  | 69.5  | 68.0  | 62.1  | 54.8  | 46.6  | 46.0  | (20) |
|                                                                        | 2%          | DB          | 40.3  | 39.5  | 48.6  | 63.9  | 75.9  | 79.1  | 83.4  | 79.9  | 68.6  | 56.9  | 47.6  | 44.6  | (21) |
|                                                                        |             | MCWB        | 39.2  | 37.8  | 41.6  | 49.0  | 60.9  | 63.3  | 67.8  | 66.1  | 59.5  | 53.6  | 46.0  | 42.9  | (22) |
|                                                                        | 5%          | DB          | 37.7  | 37.3  | 43.9  | 59.5  | 71.9  | 75.2  | 79.8  | 76.3  | 65.6  | 54.6  | 46.0  | 42.0  | (23) |
|                                                                        |             | MCWB        | 36.5  | 36.0  | 38.5  | 46.9  | 58.0  | 62.4  | 66.5  | 64.3  | 58.3  | 52.3  | 44.7  | 40.6  | (24) |
|                                                                        | 10%         | DB          | 35.7  | 35.4  | 40.5  | 54.9  | 67.6  | 71.3  | 76.4  | 72.8  | 62.9  | 52.8  | 44.3  | 39.7  | (25) |
|                                                                        |             | MCWB        | 34.8  | 34.4  | 36.9  | 44.6  | 55.7  | 59.5  | 65.1  | 62.6  | 57.0  | 50.7  | 43.1  | 38.6  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 42.2  | 40.2  | 45.6  | 53.1  | 67.4  | 68.9  | 72.2  | 71.1  | 63.4  | 56.3  | 47.9  | 46.1  | (27) |
|                                                                        |             | MCDB        | 43.0  | 41.3  | 53.9  | 64.2  | 76.3  | 78.7  | 82.3  | 79.4  | 69.2  | 58.8  | 48.7  | 47.2  | (28) |
|                                                                        | 2%          | WB          | 39.3  | 38.0  | 41.9  | 50.3  | 63.1  | 66.1  | 69.9  | 67.9  | 61.1  | 54.2  | 46.3  | 43.3  | (29) |
|                                                                        |             | MCDB        | 40.0  | 39.3  | 47.2  | 61.7  | 72.1  | 75.6  | 79.4  | 76.2  | 66.4  | 55.9  | 47.4  | 44.4  | (30) |
|                                                                        | 5%          | WB          | 36.8  | 36.1  | 39.3  | 48.0  | 60.0  | 63.5  | 68.1  | 65.9  | 59.4  | 52.8  | 44.9  | 40.7  | (31) |
|                                                                        |             | MCDB        | 37.6  | 37.1  | 42.8  | 57.2  | 68.5  | 72.2  | 76.5  | 73.3  | 64.2  | 54.4  | 45.9  | 41.7  | (32) |
|                                                                        | 10%         | WB          | 34.9  | 34.5  | 37.5  | 45.6  | 57.0  | 60.9  | 66.2  | 64.1  | 57.7  | 51.0  | 43.2  | 38.6  | (33) |
|                                                                        |             | MCDB        | 35.6  | 35.4  | 40.2  | 53.1  | 65.6  | 69.5  | 74.9  | 71.0  | 61.7  | 52.6  | 44.1  | 39.6  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 7.9   | 9.2   | 12.4  | 17.0  | 19.0  | 17.0  | 16.5  | 16.3  | 13.6  | 10.2  | 6.3   | 6.5   | (35) |
|                                                                        |             | MCDBR       | 6.3   | 7.5   | 16.5  | 25.4  | 24.6  | 23.0  | 21.6  | 22.2  | 18.1  | 11.7  | 7.2   | 6.6   | (36) |
|                                                                        | 5% WB       | MCWBR       | 6.2   | 6.1   | 10.4  | 13.2  | 11.6  | 10.2  | 8.6   | 9.7   | 10.2  | 8.6   | 6.6   | 6.3   | (37) |
|                                                                        |             | MCWBR       | 6.5   | 7.2   | 14.9  | 22.0  | 21.3  | 20.3  | 19.4  | 19.2  | 15.5  | 10.3  | 7.0   | 6.3   | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.275 | 0.305 | 0.334 | 0.370 | 0.364 | 0.364 | 0.389 | 0.386 | 0.354 | 0.334 | 0.314 | 0.269 | (40) |
|                                                                        |             | taud        | 2.129 | 2.151 | 2.215 | 2.312 | 2.387 | 2.414 | 2.362 | 2.369 | 2.444 | 2.403 | 2.222 | 2.117 | (41) |
|                                                                        | Ebn at Noon |             | 172   | 224   | 254   | 265   | 275   | 277   | 266   | 258   | 249   | 217   | 156   | 133   | (42) |
|                                                                        |             | Edh at Noon | 16    | 25    | 32    | 34    | 34    | 34    | 35    | 32    | 26    | 20    | 15    | 12    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 107   | 316   | 771   | 1257  | 1704  | 1757  | 1664  | 1319  | 837   | 386   | 120   | 68    | (44) |
|                                                                        | RadStd      |             | 23    | 60    | 98    | 140   | 173   | 129   | 159   | 132   | 96    | 54    | 18    | 11    | (45) |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (47) | Regional (38 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | -248        | N/A   | N/A   | N/A   |

Nomenclature: See separate page